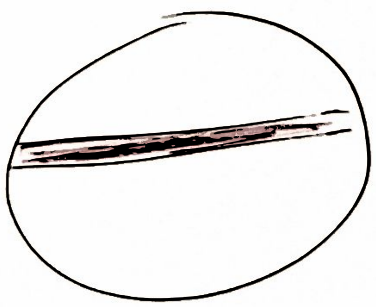
Semaphore Signal → These are movable arm signals

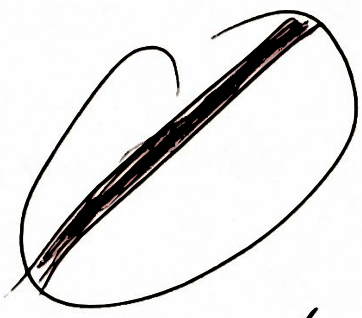
Warner signals → Defines that the train has to move @ a slower speed @ certain locations.

fixed signals are most of the time placed @ the top of the post.

Shunting signal

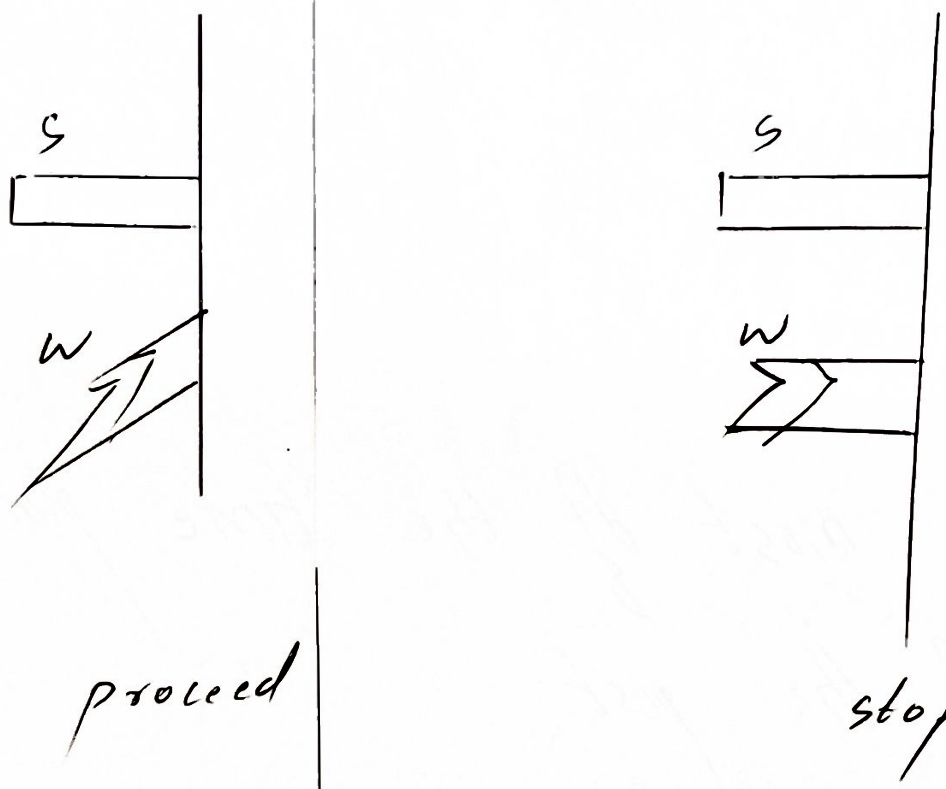


Stop



proceed

18.5) Q When semaphore & warner signals are installed in same post then for stop indicn position - both arms should be horizontal.



proceed stop

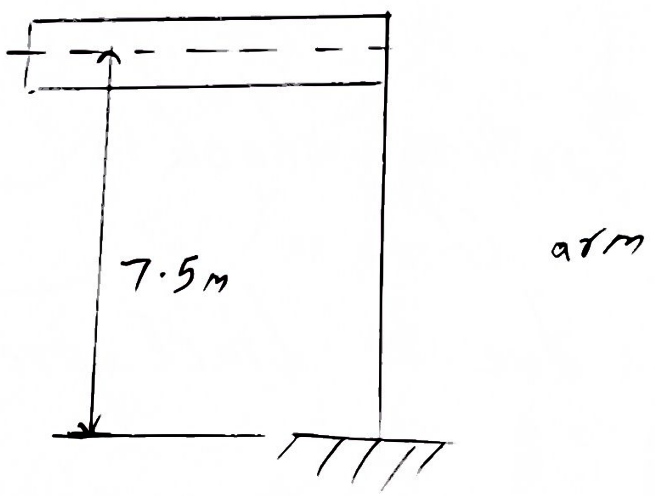
Q Yellow light hand signal → prepare or proceed cautiously

Q The reception signals are

1) Outer

2) Home signal

Q The height of centre of arm of semaphore signal above the ground is 7.5 m.



Q The treadle bar is provided - near and parallel to inner side of one of the rails.

Q The object of providing point lock is - to ensure that each switch is correctly set.

Q. Lock bar - it is a mechanical device which is used to ensure that root cannot be changed while the train is on the point even after putting back the signal.

18.7 Track resistance and power of locomotive

1) Resistance to traction.

A) Resistance due to rolling stock — Indep. of speed
   → Resistance — dep. on speed
   → atmospheric resistance

B) Resistance due to track profile

C) Resistance due to tractive effort

D) Resistance due to atmospheric / climatic condition.

A) Resistance due to rolling stock.

→ Resistance which is independent of speed

formula: 0.016 W
↳ weight of train in tonnes.

2 → Resistance dependent on speed

$RT_2 = 0.0000,8\ wV$

w → wt in tonne
V → speed of train in kmph

3 → atmospheric RT

$RT_3 = 0.000,000,6\ wV^2$

Train resistance $RT = RT_1 + RT_2 + RT_3$

Hauling capacity of locomotive:

It can be defined as the total load that can be handled by the locomotive

It indicates is an indicative of power available

16.9

HC / TR train resistance
/ traction resistance

Hauling capacity formula.

HC = μ W
↳ wn

W = total load on driving wheel

w = weight on driving axle..

n = no of pairs of driving wheels

μ = coeff of friction

Tractive effort formula.

It is taken equal to or little greater than hauling capacity.

HC ≥ T resistance

HC ≤ Trac. effort

Tractive effort : $T_e = \frac{P d^2 L}{D}$

↳ for steam locomotive

↳ diesel locomotive

$$T_e = \frac{308 (HP_f)}{V}$$

HP_f → Rated horsepower of engine

V → kmph

Tractive effort for steam loco depends on

1) Difference in pressure of cylinder (P)
2) length of stroke (L)
3) Diameter of piston (d)
4) Area of piston (A)
5) Diameter of wheel (D)

18.11

Airport

Airport: it is an area regularly used for landing & takeoff of aircrafts.

Airport is also provided with facilities like shelter, repair of aircraft & for passengers boarding, deboarding & cargo boarding-deboarding.

Runway orientation:

The orientation of runway depends on the direction of wind & to some extent area available for developement

Runway is oriented in the direction of prevailing winds such that the direction of wind is opposite to the direction of landing & takeoff.

Runway orientation is done by graphical method — wind Rose diagram.

In wind rose diagram, wind data is means prepared by direction, duration & intensity of wind.

At least 5 years wind Rose data is to be considered & preferably 10 years data is considered for design.

- BRL (basic runway length)
- Basic runway length is increased @ the rate of 7:1 per 300 m rise in elevation above mean sea level.

10 11
18·13

BRL is increased by 1% for 1°C rise in temperature above standard atmospheric temperature.

BRL is increased by 20% for every 1% of effective gradient.

Taxiway (turning radius)

Main function of taxiway is to provide linkage to the aircraft from the runway to loading apron or service hangar.

hangar: it is the covered area for repair & servicing of aircraft (1814)

number of hangars depends on peak hour volume of aircraft & also depends on size of aircraft & turning radius.

Turning radius = to change the direction of aircraft path a horizontal curve is provided.

$$R = \frac{V^2}{125 f}$$

V → kmph

f → transverse friction coefficient 0.13.

R → m

Apron: paved area for aircraft parking, loading & unloading of passengers & cargoes.

The size of apron depends on number of gate positions. (waiting capacity for aircraft)

formula

$$\text{no of gate posns} = \frac{\text{capacity of runway} \times \text{Avg gate occupancy time}}{60}$$

Exit taxiway

taxiway should be located such that the landing aircraft leaves the runway as early as possible such taxiway is called exit taxiway.

Apron turnlable

it is used to expedite the parking of aircraft & save parking space.

Control tower

Visibility all around and sound proof chamber

The size of hangar in an airport is guided by -

The length, height & wing span of aircraft.

Taxiway design depends on following factors

1 volume of air traffic

2 runway configuration

3 location of terminal building, loca of hangar

18.17

full forms

A.T.C : Air traffic control

V.F.R : visual flight rule

L.F/M.F ratio range = low freq
/medium freq

I L S = Instrumental landing system.

IFR : Instrument fly rule

Colours

Runway marking = white paint

taxiway marking = yellow paint.

all runway side lights : white

all taxiway side lights = blue

factors affecting the runway capacity

1. ATC measures
2. VFR/IFR operation
3. Runway configuration
4. Aircraft traffic composition.

ICAO : airport code.

Instrument installed in sequencing of landing Operation

1) Z marker
2) threshold light
3) Glide slope antenna.
4) localiser antenna.

Harbour classification

Based on geographical genesis,

1) Natural harbour

Natural harbours - Storms & waves are protected by natural land contours. Eg Bombai

2) Semi natural harbour

When a harbour is protected one side from natural land contour and other side man made construction. Eg Vishakha pattnam

3) Artificial harbour

Eg chennai port.

27 Aug 2024 Class 19

Heliport / Helipad :

The place where designed area provided for takeoff and landing of Helicopter.

VTOL = Vertical takeoff & landing

Dock and harbour :

It is the place designated for ship or boat or water bodies.

A Harbour is a protected water area to provide safe & suitable accomodation for vessels seeking supplies refuling refuge, repair or transfer of passengers and cargo.

Definitions

2000 purpose & name

1) Jetties: These are structures in the form of piled projections and they are built out from the shore to deep water.

It may be constructed either for a navigable river or in the sea.

2) Wharf & quays: These are constru-cted parallel to the shore with the harbour to permit berthing of vessel it gives necessary depth for navigation.

Based on usage

1) fishery harbour
2) Military harbour
3) Refugee harbour
4) Commercial harbour.

Based on site selection

1) Canal harbour Eg Panal PR
2) lake harbour
3) River harbour
4) Sea harbour

When built along L // to shore then it is called quays.

Fender or dolphin: These are the piles provided in water front areas.

Dock is an artificial enclosure for reception of ship.

Tunneling

A tunnel is an underground passage completely enclosed except for opening for exit, commonly @ each end.

Operations involved in tunneling

- Drilling & blasting
  - Grouting
    - Excavation
      - Supporting
        - transportation of muk
          - lining or coating
            - Draining
              - Ventil"

19 7 TBM: tunnel boring machine.

Facet points :-

1. In full face method of constructing tunnel, the first involved is — excvation being done along the perimeter.

2. Shape of tunnel	type of rock.
a) Horse shoe shape	Soft rock
b) Circular section 1997	Water carrying [best for non cohesive soil] structure
c) Egg shaped tunnel 1998	Sewer (to maintain self cleaning velocity)
d) Segmental roof section	Subway

3. floating dock:

A ship is berthed in a chamber and lifted by principle of buoyancy such structures are called as floating dock.

4. Wind rose diagram is used for :-

runway orientation

Purpose and name

1)	Breakwater	protects a sea shore
2)	Wharf	lays vessel along side, recieves, discharges cargo & passenger
3)	fender	to absorb energy of moving waves. / vessels.
4)	Revetments	Separate the land from seawater.

year 2000

19.7

Method of tunneling adopted for soft soil / rock = Needle beam method.

Hard rock ke liye = Heading & Benching method.

Airport terms & uses / purpose: Year 2000

1) PNB → Noise level.

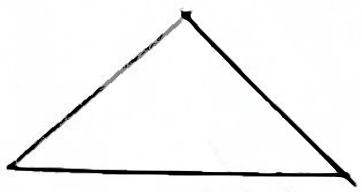
2) LCN → Airport pavement evaluation

3) STOL → Type of aircraft.

4) ICAO →

Q2000 Tunnel alignment is carried out by

→ triangulation.



Q.2001 In deep water waves individual particles

- do not rotate in vertical plane about its horizontal axis.

Q2004

Name	Purpose
① Dynamometer car	gives information about the condition of railway track.
② Turntable	Arrangement for turning the direction of engine.
③ Crossing Station	Station on a single railway track system where up & down trains can ~~cross~~ pass each other.

19.11.2005

Q Use of station pointer:

For making tidal observations.

Q Moorings of ship: anchoring of ship.

Q. Tunnel Ventilation - best method: combination of blow in & blow out method

Q. Which of the statements is correct for turnout:

1) ✓ in diamond interchange there is possibility of illegal wrong way turns.

19.12

2) X Diamond interchange is far superior than cloverleaf interchange design.

Q. 1995 What is the number of crossings if the angle formed between running gauge faces is A. (a) tan A (b) sec A (c) cot A (d) A rad

↳ Ans (c) cot A.

Q 1997 What will be the curve lead for 1 in 8.5 turnout taking off from a straight broad gauge track.

curve lead : 2GN

BG : 1.676 m

N : 8.5

CL : 2 × 1.676 × 8.5 : 28.5 m.

19.13| Q[1997] Composition of rail steel = CSPMnSi

C + Mn + Si + S + P

सी एम एस पी

Phosphorus

Q. Ordinary rail is made by high carbon steel.

Q Rails are made by open hearth process.

Q What is the value of grade compensation on 4° curve in broad gauge railway track.

BG → 4° → GC = ?

0.4% × 4

= 0.16%.

Q. Cut & cover method is used in construction of metro railways.

for :- Shallow tunnel construction

When the soil is loose soft soil and water table is high. In this trench is excavated and roofed over with a overhead support system.

Q versine of curve = setback = $\frac{L^2}{8R}$

Instrument	Purpose
a. Rail tong →	to lift rail
Shovel →	To handle ballast
claw bar ←	To remove dog spikes out of sleeper
Crowbar ←	To correct track alignment.